EXERCISE-I (Conceptual Questions) Build Up Your Understanding

RATE OF REACTION

1. Consider the chemical reaction :
$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$
The rate of this reaction can be expressed in terms of concentration of $N_2(g)$, $H_2(g)$ or $NH_3(g)$. Identify the correct relationship amongst the rate expressions.

(1) Rate $= -\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

(2) Rate $= -\frac{d[N_2]}{dt} = -\frac{3d[H_2]}{dt} = \frac{2d[NH_3]}{dt}$

(3) Rate $= \frac{d[N_2]}{dt} = \frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

(4) Rate $= -\frac{d[N_2]}{dt} = -\frac{d[H_2]}{dt} = \frac{d[NH_3]}{dt}$

CK0001

2. In the formation of sulphur trioxide by the contact process $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$; the rate of reaction is expressed as

$$-\frac{d[O_2]}{dt} = 2.5 \times 10^{-4} \text{ mol } L^{-1} s^{-1}$$

The rate of disappearance of (SO_2) will be
(1) 5×10^{-4} mol L^{-1} s^{-1}
(2) -2.25×10^{-4} mol L^{-1} s^{-1}
(3) 3.75×10^{-4} mol L^{-1} s^{-1}
(4) 50.0×10^{-4} mol L^{-1} s^{-1}

CK0002

3. In a catalytic reaction involving the formation of ammonia by Haber's process $N_2 + 3H_2 \rightarrow 2NH_3$ the rate of appearance of NH_3 was measured as 2.5×10^{-4} mole L^{-1} s^{-1} The rate of disappearance of H_2 will be –
(1) 2.5×10^{-4} mol L^{-1} s^{-1}
(2) 1.25×10^{-4} mol L^{-1} s^{-1}
(3) 3.75×10^{-4} mol L^{-1} s^{-1}
(4) 5×10^{-4} mol L^{-1} s^{-1}

CK0003

4. Which of the following statement is correct for a reaction X + 2Y → Product (p)
(1) The rate of disappearance of X = twice the rate of disappearance of Y.
(2) The rate of disappearance of X = ½ rate of appearance of product (p)
(3) The rate of appearance of product (p) = ½the rate of disappearance of Y
(4) The rate of appearance of product (p) = ½the rate of disappearance of X

CK0004

5. For the reaction, $N_2O_5 \longrightarrow 2NO_2 + \frac{1}{2}O_2$

Given $-\frac{d[N_2O_5]}{dt} = K_1[N_2O_5]$, $\frac{d[NO_2]}{dt} = K_2[N_2O_5]$

$\frac{d[O_2]}{dt} = K_3[N_2O_5]$

The relation between K_1, K_2 and K_3 is –
(1) $2K_1 = K_2 = 4K_3$ (2) $K_1 = K_2 = K_3$
(3) $2K_1 = 4K_2 = K_3$ (4) None

CK0005

6. Rate of formation of SO_3 according to the reaction $2SO_2 + O_2 \rightarrow 2SO_3$ is 1.6×10^{-3} kg L^{-1} min^{-1} Hence rate of decomposition of SO_2 is :–
(1) 1.6×10^{-3} kg L^{-1} min^{-1}
(2) 8.0×10^{-4} kg L^{-1} min^{-1}
(3) 3.2×10^{-3} kg L^{-1} min^{-1}
(4) 1.28×10^{-3} kg L^{-1} min^{-1}

CK0006

7. For a general chemical change 2A+3B → products, the rate of disappearance of A is r_1 and of B is r_2. The rates r_1 and r_2 are related as :–
(1) $3r_1 = 2r_2$ (2) $r_1 = r_2$
(3) $2r_1 = 3r_2$ (4) $r_1^2 = 2r_2^2$

CK0007

8. For the reaction
$2A + B \rightarrow 3C + D$
Which of the following does not express the reaction rate ?

(1) $-\frac{1}{3}\frac{d[C]}{dt}$ (2) $-\frac{d[B]}{dt}$ (3) $\frac{d[D]}{dt}$ (4) $-\frac{1}{2}\frac{d[A]}{dt}$

CK0117

9. Consider the reaction
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

The correct equality relationship between $\frac{d[NH_3]}{dt}$ and $-\frac{d[H_2]}{dt}$ is :

(1) $\frac{d[NH_3]}{dt} = \frac{1}{3} - \frac{d[H_2]}{dt}$

(2) $+\frac{d[NH_3]}{dt} = \frac{2}{3} - \frac{d[H_2]}{dt}$

(3) $+\frac{d[NH_3]}{dt} = \frac{3}{2} - \frac{d[H_2]}{dt}$

(4) $\frac{d[NH_3]}{dt} = -\frac{d[H_2]}{dt}$

CK0118

RATE LAW / ORDER / MOLECULARITY

10. For gaseous reaction, rate = k [A] [B]. If volume of container is reduced to $\frac{1}{4}$ of initial, then the rate of the reaction will be..... times of initial:-

(1) $\frac{1}{8}$ (2) 8 (3) $\frac{1}{16}$ (4) 16

CK0010

11. $2A_{(g)} + B_{(g)} \longrightarrow$ Product is an elementary reaction, If pressure of both A and B is increased three times of the initial pressure, the rate of forward reaction will be ------ of the previous rate :-

(1) 9 times (2) 27 times

(3) $\frac{1}{9}$ times (4) $\frac{1}{27}$ times

CK0011

12. The rate of certain hypothetical reaction A + B + C → products is given by

$r=\frac{-d[A]}{dt}=K[A]^{\frac{1}{2}}[B]^{\frac{1}{3}}[C]^{\frac{1}{4}}$ The order of the reaction –

(1) 1 (2) $\frac{1}{2}$ (3) 2 (4) $\frac{13}{12}$

CK0012

13. Which of the following rate law has an overall order of 0.5 for reaction involving substances x,y and z ?

(1) Rate = K (C_x) (C_y) (C_z)

(2) Rate = K $(C_x)^{0.5}(C_y)^{0.5}(C_z)^{0.5}$

(3) Rate = K $(C_x)^{1.5}$ $(C_y)^{-1}(C_z)^{\circ}$

(4) Rate = K$(C_x)(C_z)^{\circ}$ / $(C_y)^2$

CK0013

14. A chemical reaction involves two reacting species. The rate of reaction is directly proportional to square of the concentration of one of them and inversely proportional to the concentration of the other. The order of reaction is –

(1) 1 (2) 2

(3) Zero (4) Unpredictable

CK0014

15. For the reaction $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$, the experimental data suggests, Rate = K $[H_2]$ $[Br_2]^{\frac{1}{2}}$. The order for this reaction is –

(1) 2 (2) $1\frac{1}{2}$ (3) 1 (4) $2\frac{1}{2}$

CK0015

16. Select the rate law that corresponds to the data shown for the following reaction A + B → C

Exp.	[A]	[B]	Initial rate
1.	0.012	0.035	0.10
2.	0.024	0.070	1.6
3.	0.024	0.035	0.20
4.	0.012	0.070	0.80

(1) Rate = K $[B]^3$ (2) Rate = K$[B]^4$

(3) Rate = K[A]$[B]^3$ (4) Rate = K$[A]^2[B]^2$

CK0016

17. Select the law that corresponds to data shown for the following reaction 2A + B → C + D :-

Exp.	[A]	[B]	Initial rate ($mol\,L^{-1}\,min^{-1}$)
1.	0.1	0.1	7.5×10^{-3}
2.	0.3	0.2	9.0×10^{-2}
3.	0.3	0.4	3.6×10^{-1}
4.	0.4	0.1	3.0×10^{-2}

(1) Rate = K $[A]^2$[B] (2) Rate = K[A]$[B]^2$

(3) Rate = K[A]$[B]^3$ (4) Rate = K[A][B]

CK0018

18. For a hypothetical reaction ; A + B → C the following data were obtained in three different experiments :–

[A] ($mol\,L^{-1}$)	[B] ($mol\,L^{-1}$)	Rate of reaction ($mol\,L^{-1}\,min^{-1}$)
0.01	0.01	1.0×10^{-4}
0.01	0.03	9.0×10^{-4}
0.03	0.03	2.70×10^{-3}

The rate law will be :–

(1) r = K$[A]^2$[B] (2) r = K[A]$[B]^2$

(3) r = K[A][B] (4) None of these

CK0019

19. Calculate the order of the reaction w.r.t. A and B:

[A] ($mol\,L^{-1}$)	[B] ($mol\,L^{-1}$)	Rate ($mol\,L^{-1}S^{-1}$)
0.05	0.05	1.2×10^{-3}
0.10	0.05	2.4×10^{-3}
0.05	0.10	1.2×10^{-3}

(1) 1 and 0 (2) 1 and 1

(3) 0 and 1 (4) None

CK0020

20. For a chemical reaction A + B → product, the order is one with respect to each A and B. Value of x and y from the given data is :–

Rate ($mol\ L^{-1}s^{-1}$)	[A]	[B]
0.10	0.20 M	0.05 M
0.40	x	0.05 M
0.80	0.40 M	y

(1) 0.20, 0.80 (2) 0.80, 0.40

(3) 0.80, 0.20 (4) 0.40, 0.20

CK0021

21. The rate law for the single step reaction 2A + B → 2C, is given by –

(1) Rate = K[A][B] (2) Rate = $K[A]^2[B]$

(3) Rate = K[2A][B] (4) Rate = $K[A]^2[B]^\circ$

CK0023

22. For a reaction of the type A + B → products, it is observed that doubling the concentration of A causes the reaction rate to be four times as great, but doubling the amount of B does not effect the rate. The rate equation is –

(1) Rate = K [A][B] (2) Rate = K $[A]^2$

(3) Rate = $K[A]^2[B]$ (4) Rate = $K[A]^2[B]^2$

CK0024

23. Point out incorrect statement.

(1) Rate law is an experimental value

(2) Law of mass action is a theoretical proposal

(3) Rate law is more informative than law of mass action for developing mechanism

(4) Rate law is always different from the expression of law of mass action.

CK0025

24. For an elementary process 2X + Y → Z + W, the molecularity is –

(1) 2 (2) 1

(3) 3 (4) Unpredictable

CK0026

25. For a reaction A + B → products, the rate of the reaction was doubled when the concentration of A was doubled, the rate was again doubled when the concentration of both A & B were doubled the order of the reaction with respect to A & B are:–

(1) 1, 1 (2) 2, 0 (3) 1, 0 (4) 0, 1

CK0027

26. For the reaction A + B → products, it is found that the order of A is 1 and the order of B is $\frac{1}{2}$. When the concentration of both A and B are increased four times, the rate will increase by a factor of :–

(1) 16 (2) 8 (3) 6 (4) 4

CK0028

27. The rate law for a reaction A + B → product is rate = $K[A]^1[B]^2$. Then which one of the following statement is false :–

(1) If [B] is held constant while [A] is doubled, the reaction will proceed twice as fast

(2) If [A] is held constant while [B] is reduced to one quarter, the rate will be halved

(3) If [A] and [B] are both doubled, the reaction will proceed 8 times as fast

(4) This is a third order reaction

CK0029

28. For a chemical reaction A → B, the rate of reaction doubles when the concentration of A is increased 8 times. The order of reaction w.r.t. A is :–

(1) 3 (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) Zero

CK0030

29. The rate constant of a first order reaction depends on the :–

(1) Concentration of the reactant

(2) Concentration of the product

(3) Time

(4) Temperature

CK0031

30. Consider the following reaction,

$KClO_3 + 6FeSO_4 + 3H_2SO_4 \longrightarrow KCl + 3Fe_2(SO_4)_3 + 3H_2O$

Which of the statements regarding this reaction is incorrect?

(1) It is a complex reaction

(2) It is a tenth order reaction

(3) It is completed in several steps

(4) Order of this reaction can be determined experimentally

CK0215

PSEUDO FIRST ORDER REACTION / MECHANISM OF REACTION

31. Following mechanism has been proposed for a reaction 2A + B → D + E

A + B → C + D .....(slow)

A + C → E .....(fast)

The rate law expression for the reaction is –

(1) r = $K[A]^2[B]$ (2) r = K[A][B]

(3) r = $K[A]^2$ (4) r = K[A][C]

CK0032

32. The chemical reaction $2O_3 \rightarrow 3O_2$ proceeds as follows

$O_3 \rightleftharpoons O_2 + O$ ..... (fast)

$O + O_3 \rightarrow 2O_2$ ..... (slow)

The rate law expression should be –

(1) $r = K[O_3]^2$ (2) $r = K[O_3]^2[O_2]^{-1}$

(3) $r = K[O_3][O_2]$ (4) Unpredictable

CK0033

33. The hypothetical reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below –

$A_2 \rightleftharpoons A + A$ ..... (fast)

$A + B_2 \rightarrow AB + B$ ..... (slow)

$A + B \rightarrow AB$ ..... (fast)

The order of the over all reaction is –

(1) 2 (2) 1 (3) 1½ (4) Zero

CK0034

34. The rate for the reaction

RCl + NaOH (aq) → ROH + NaCl is given by rate = $k_1[RCl]$. The rate of the reaction is –

(1) Doubled on doubling the concentration of NaOH

(2) Halved on reducing the concentration of RCl to half

(3) Decreased on increasing the temperature of reaction

(4) Unaffected by increasing the temperature of the reaction

CK0035

35. For reaction $NO_2 + CO \rightarrow CO_2 + NO$, the rate expression is, Rate = $k[NO_2]^2$

The number of molecules of CO involved in the slowest step will be –

(1) 0 (2) 1 (3) 2 (4) 3

CK0036

36. The rate law of the reaction A + 2B → product is given by $\frac{d[P]}{dt} = K[A]^2[B]$. If A is taken in large excess, the order of the reaction will be –

(1) Zero (2) 1 (3) 2 (4) 3

CK0037

37. The acidic hydrolysis of ester is –

(1) Psuedo Ist order reaction

(2) IInd order reaction

(3) Trimolecular reaction

(4) All of the above

CK0038

38. The reaction 2A + B → P, follows the mechanism

$2A \rightleftharpoons A_2$ (fast)

$A_2 + B \rightarrow P$ (slow)

The order of the reaction is :–

(1) 1.5 (2) 3 (3) 1 (4) 2

CK0041

39. For the reaction $2NO + Cl_2 \rightarrow 2NOCl$ the following mechanism has been proposed

$NO + Cl_2 \rightleftharpoons NOCl_2$ (fast)

$NOCl_2 + NO \rightarrow 2NOCl$ (slow) the rate law for the reaction is :-

(1) Rate = $K[NO]^2[Cl_2]$ (2) Rate = $K[NO][Cl_2]^2$

(3) Rate = $K[NOCl_2]$ (4) Rate = $K[NOCl]^2$

CK0042

ZERO / FIRST / nth ORDER REACTIONS

40. The rate constant is numerically the same for three reactions of first, second and third order respectively. Which one is true at a moment for rate of all three reactions if concentration of reactants is same and greater than 1 M.

(1) $r_1 = r_2 = r_3$ (2) $r_1 > r_2 > r_3$

(3) $r_1 < r_2 < r_3$ (4) All

CK0043

41. K for a zero order reaction is 2×10^{-2} mol L^{-1} s^{-1}. If the concentration of the reactant after 25 s is 0.5 M, the initial concentration must have been.

(1) 0.5 M (2) 1.25 M

(3) 12.5 M (4) 1.0 M

CK0044

42. The decomposition of N_2O_5 occurs as, $2N_2O_5 \rightarrow 4NO_2 + O_2$, and follows first order kinetics; hence

(1) The reaction is bimolecular

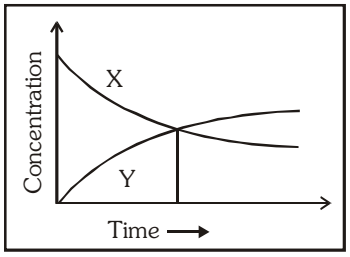
(2) The recation is unimolcular

(3) $t_{1/2}$ α a°

(4) $t_{1/2}$ α a^2

CK0045

43. The accompanying figure depicts the change in concentration of species X and Y for the reaction X → Y as a function of time the point of intersection of the two curves reperesents.



(1) $t_{1/2}$

(2) $t_{3/4}$

(3) $t_{2/3}$

(4) Data are insufficient to predict

CK0046

44. If the first order reaction involves gaseous reactants and gaseous products the unit of its rate is –
(1) atm. (2) atm.s
(3) $atm.s^{-1}$ (4) $atm^2\ s^2$
CK0047

45. Plot of log(a – x) vs time t is straight line. This indicates that the reaction is of –
(1) Second order (2) First order
(3) Zero order (4) third order
CK0048

46. The rate constant of a first order reaction is $4 \times 10^{-3}\ s^{-1}$. At a reactant concentration of 0.02 M, the rate of reaction would be–
(1) $8 \times 10^{-5}\ M\ s^{-1}$ (2) $4 \times 10^{-3}\ M\ s^{-1}$
(3) $2 \times 10^{-1}\ M\ s^{-1}$ (4) $4 \times 10^{-1}\ M\ s^{-1}$
CK0049

47. In a first order reaction the concentration of the reactant is decreased from 1.0 M to 0.25 M in 20 min. The rate constant of the reaction would be–
(1) $10\ min^{-1}$ (2) $6.931\ min^{-1}$
(3) $0.6931\ min^{-1}$ (4) $0.06931\ min^{-1}$
CK0050

48. In a first order reaction the a/(a – x) was found to be 8 after 10 minute. The rate constant is
(1) $\frac{(2.303 \times 3\log 2)}{10}$
(2) $\frac{(2.303 \times 2\log 3)}{10}$
(3) $10 \times 2.303 \times 2\log 3$
(4) $10 \times 2.303 \times 3\log 2$
CK0051

49. 75 % of a first order reaction was found to complete in 32 min. When will 50 % of the same reaction complete –
(1) 24 min (2) 16 min
(3) 8 min (4) 4 min
CK0052

50. A first order reaction has a half life period of 69.3 s. At 0.10 mol L^{-1} reactant concentration, the rate will be –
(1) $10^{-4}\ M\ s^{-1}$ (2) $10^{-3}\ M\ s^{-1}$
(3) $10^{-1}\ M\ s^{-1}$ (4) $6.93 \times 10^{-1}\ M\ s^{-1}$
CK0053

51. For a first order reaction A → products, the rate of reaction at [A] = 0.2 M is 1×10^{-2} mol $L^{-1}min^{-1}$. The half life period for the reaction is –
(1) 832 min. (2) 440 s
(3) 416 min. (4) 14 min.
CK0054

52. 99 % of a first order reaction was completed in 32 min. when will 99.9 % of the reaction complete ?
(1) 50 min. (2) 46 min.
(3) 49 min. (4) 48 min.
CK0055

53. What is the half life of a radioactive substance if 87.5% of any given amount of the substance disintegrate in 40 minutes ?
(1) 160 min (2) 10 min
(3) 20 min (4) 13 min 20 sec.
CK0057

54. For a given reaction of first order it takes 20 minute for the concentration to drop from 1 M to 0.6 M. The time required for the concentration to drop from 0.6 M to 0.36 M will be :
(1) More than 20 min
(2) Less than 20 min
(3) Equal to 20 min
(4) Infinity
CK0058

55. A first order reaction is carried out with an initial concentration of 10 mol per litre and 80% of the reactant changes into the product. Now if the same reaction is carried out with an initial concentration of 5 mol per litre for the same period the percentage of the reactant changing to the product is.
(1) 40 (2) 80
(3) 160 (4) Cannot be calculated
CK0059

56. The reaction L → M is started with 10 g/L. After 30 minute and 90 minute, 5 g/L and 1.25 g/L are left respectively. The order of reaction is
(1) 0 (2) 2
(3) 1 (4) 3
CK0061

57. Doubling the initial concentration of a reactant doubles $t_{1/2}$ of the reaction, then order of the reaction is–
(1) 3 (2) 2
(3) 1 (4) 0
CK0062

58. The half life period for catalytic decomposition of AB_3 at 50 mm is found to be 4 hrs and at 100 mm it is 2 hrs. The order of reaction is –
(1) 3 (2) 1 (3) 2 (4) 0
CK0063

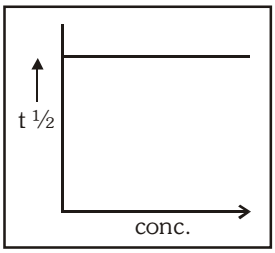
59. The rate constant for a reaction is 10.8×10^{-5} mol L^{-1} s^{-1} The reaction obeys –
(1) First order (2) Zero order
(3) Second order (4) All are wrong
CK0064

60. A substance 'A' decomposes in solution following the first order kinetics flask I contains 1 L of 1M. solution of A and flask II contains. 100 ml of 0.6 M solution. After 8 hr. the concentration of A in flask. I become 0.25 M, what will be the time for concetration of A in flask II to become 0.3 M.
(1) 0.4 hr.
(2) 2.4 hr.
(3) 4.0 hr.
(4) Unpredictable as rate constant is not given
CK0065

61. The rate constant (K) for the reaction $2A + B \rightarrow$ product, was found to be 2.57×10^{-5} L mol^{-1} s^{-1} after 15 s, 2.60×10^{-5} L mol^{-1} s^{-1} after 30 s and 2.55×10^{-5} L mol^{-1} s^{-1} after 50 s. The order of reaction is
(1) 2 (2) 3 (3) Zero (4) 1
CK0066

62. The rate constant for a second order reaction is 8×10^{-5} M^{-1} min^{-1} : How long will it take a 1M solution to be reduced to 0.5 M.
(1) 8.665×10^3 min (2) 8×10^{-3} min
(3) 1.25×10^4 min (4) 4×10^{-5} min
CK0067

63. A graph between $t_{1/2}$ and concentration for n^{th} order reaction is a straight line. Reaction of this nature is completed 50% in 10 minutes when concentration is 2 mol L^{-1}. This is decomposed 50% in t minutes at 4 mol L^{-1}, n and t are respectively



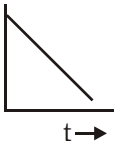
(1) 0, 20 min. (2) 1, 10 min.
(3) 1, 20 min. (4) 0, 5 min.
CK0068

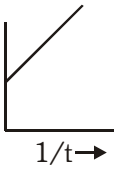
64. In the case of first order reaction, the ratio of time required for 99.9 % completion to 50 % completion is :–
(1) 2 (2) 5 (3) 10 (4) None
CK0070

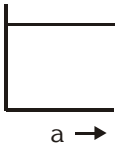
65. From different sets of data of $t_{1/2}$ at different initial concentrations say 'a' for a given reaction, the $[t_{1/2} \times a]$ is found to be constant. The order of reaction is :–
(1) 0 (2) 1 (3) 2 (4) 3
CK0071

66. Which of the following statement is not correct for the reaction whose rate is r = k (rate constant)
(1) rate of reaction is independent of concentration of reactant
(2) $t_{1/2}$ of reaction does not depend upon concentration of reactant
(3) rate constant is independent of concentration of reactant
(4) this is zero order reaction
CK0073

67. Which of the following curves represents a I^{st} order reaction :–

(1) 

(2) 

(3) 

(4) 1 & 3 both
CK0074

68. The following data were obtained at a certain temperature for the decomposition of ammonia

p (mm)	50	100	200
$t_{1/2}$	3.64	1.82	0.91

The order of the reaction is :–
(1) 0 (2) 1
(3) 2 (4) 3
CK0075

69. A reaction is found to have the rate constant x s^{-1} , by what factor the rate is increased if initial concentration of reactant is tripled
(1) 3
(2) 9
(3) x
(4) Remains same
CK0076

70. Which is incorrect :–
(1) Half life of a first order reaction is independent of initial concentration
(2) Rate of reaction is constant for first order reaction
(3) Unit of K for second order reaction is mol^{-1} L s^{-1}
(4) Half life of zero order is proportional to initial concentration

CK0077

71. Hydrolysis of ester in alkaline medium is :–
(1) First order reaction
(2) Second order reaction
(3) Zero order reaction
(4) Third order reaction

CK0078

72. The expression which gives $\frac{1}{4}$th life of I[st] order reaction is :–
(1) $\frac{K}{2.303}\log\frac{4}{3}$ (2) $\frac{2.303}{K}\log 3$
(3) $\frac{2.303}{K}\log\frac{3}{4}$ (4) $\frac{2.303}{K}\log\frac{4}{3}$

CK0079

73. The rate constant of a zero order reaction is 0.2 mol $dm^{-3}h^{-1}$. If the concentration of the reactant after 30 minutes is 0.05 mol dm^{-3}. Then its initial concentration would be :-
(1) 6.05 mol dm^{-3} (2) 0.15 mol dm^{-3}
(3) 0.25 mol dm^{-3} (4) 4.00 mol dm^{-3}

CK0080

74. A reaction is of first order. After 100 minutes 75 gm of the reactant A are decomposed when 100 gm are taken initially, calculate the time required when 150 gm of the reactant A are decomposed, the initial weight taken is 200 gm :-
(1) 100 minutes (2) 200 minutes
(3) 150 minutes (4) 175 minutes

CK0081

75. For which of the following, the unit of rate and rate constant of the reaction are identical :-
(1) First order reaction
(2) Zero order reaction
(3) Second order reaction
(4) Fractional order of reaction

CK0082

76. $A \rightarrow$ Product, follows the first order kinetics. If the half life period of the reaction at $[A]_{Initial}$ = 0.2 mol L^{-1} is 20 min. then the value of rate constant would be:-
(1) 4 s^{-1} (2) 20 s^{-1}
(3) 4 lit. mol^{-1} $min.^{-1}$ (4) 0.034 min^{-1}

CK0083

77. Correct statement about first order reaction is:-
(1) $t_{completion}$ = finite
(2) $t_{1/2} \propto \frac{1}{a}$
(3) Unit of K is mole lit^{-1} s^{-1}
(4) $t_{1/2} \times K$ = constant (at given temperature)

CK0084

COLLISION THEORY AND FACTORS AFFECTING RATE OF REACTION

78. Activation energy of a reaction is –
(1) The energy released during the reaction
(2) The energy evolved when activated complex is formed
(3) Minimum extra amount of energy needed to overcome the potential barrier of reaction
(4) The energy needed to form one mole of the product

CK0086

79. The minimum energy for molecules to enter into chemical reaction is called.
(1) Kinetic energy (2) Potential energy
(3) Threshold energy (4) Activation energy

CK0087

80. The rate constant k_1 of a reaction is found to be double that of rate constant k_2 of another reaction. The relationship between corresponding activation energies of the two reactions at same temperature (E_1 & E_2) can be represented as
(1) $E_1 > E_2$ (2) $E_1 < E_2$
(3) $E_1 = E_2$ (4) $E_1 = 4E_2$

CK0088

81. At room temperature the reaction between NO and O_2 to give NO_2 is fast while that between CO and O_2 is slow it is due to –
(1) CO is smaller in size than that of NO
(2) CO is poisonous
(3) The activation energy for the reaction $2NO + O_2 \rightarrow 2NO_2$ is less than $2CO + O_2 \rightarrow 2CO_2$
(4) NO_2 is poisonous

CK0089

82. Chemical reaction occurs as a result of collision between reacting molecules. Therefore the reaction rate is given by
(1) Total number of collisions occuring in a unit volume per second
(2) Fraction of molecules which possess energy less than the threshold energy.
(3) Total number of effective collisions
(4) Temperature

CK0090

83. An endothermic reaction A → B have an activation energy 15 kCal/mol and the heat of the reaction is 5 k cal/mol. The activation energy of the reaction B → A is –
(1) 20 kCal/mol (2) 15 kCal/mol
(3) 10 kCal/mol (4) Zero
CK0091

84. A large increase in the rate of a reaction for a rise in temperature is due to –
(1) Increase in the number of collisions
(2) Increase in the number of activated molecules
(3) Lowering of activation energy
(4) Shortening of the mean free path
CK0092

85. Rate of which reactions increases with temperature
(1) Of any reaction
(2) Of exothermic reaction
(3) Of endothermic reaction
(4) Can't be predicted
CK0093

86. The rate of a chemical reaction doubles for every 10^0C rise in temperature. If the temp is increased by 60^0C the rate of reaction increases by :
(1) 20 times (2) 32 times
(3) 64 times (4) 128 times
CK0094

87. According to the arrhenius equation a straight line is to be obtained by plotting the logarithm of the rate constant of chemical reaction (log k) against.
(1) (T) (2) log T
(3) $\frac{1}{T}$ (4) log $\frac{1}{T}$
CK0095

88. Which plot will give the value of activation energy.
(1) k v/s T (2) $\frac{1}{k}$ v/s T
(3) lnk v/s T (4) lnk v/s $\frac{1}{T}$
CK0096

89. Given that k is the rate constant for some order of any reaction at temp T then the value of $\lim_{T\to\infty} \log k$ _____.
(1) A/2.303 (2) A
(3) 2.303 A (4) log A
CK0097

90. The rate constant, the activation energy and the Frequency factor of a chemical reaction at 25^0C are 3×10^{-4} s^{-1}; 104.4 kJ mol^{-1} and 6.0×10^{14} s^{-1} respectively, the value of the rate constant as $T \to \infty$ is.
(1) 2×10^8 s^{-1} (2) 6×10^{14} s^{-1}
(3) Infinity (4) 3.6×10^{30} s^{-1}
CK0099

91. For an endothermic reaction where ΔH represents the enthalpy of the reaction in kJ/mol ; the minimum value for the energy of activation will be
(1) Less than ΔH (2) Zero
(3) More than ΔH (4) Equal to ΔH
CK0100

92. The rate of reaction increases to 2.3 times when the temperature is raised from 300 K to 310 K. If K is the rate constant at 300 K then the rate constant at 310 K will be equal to –
(1) 2 k (2) k
(3) 2.3 k (4) 3 k^2
CK0101

93. If the concentration units are reduced by n times then the value of rate constant of first order will
(1) Increases by n times
(2) Decreases by factor of n
(3) Remain constant
(4) Decrease 1/n times
CK0103

94. Which is used in the determination of reaction rates.
(1) Reaction Temperature
(2) Reaction Concentration
(3) Specific rate constant
(4) All of these
CK0104

95. The rate constant of a first order reaction depends on the :–
(1) Concentration of the reactant
(2) Concentration of the product
(3) Volume of the reaction container
(4) Temperature
CK0105

96. For the decomposition of $N_2O_5(g)$ it is given that– $2N_2O_5(g) \to 4NO_2(g) + O_2(g)$ activation energy = E_a

$N_2O_5(g) \to 2NO_2(g) + \frac{1}{2}O_2(g)$ activation energy = E'_a

then
(1) $E_a = 2E'_a$ (2) $E_a > E'_a$
(3) $E_a < E'_a$ (4) $E_a = E'_a$
CK0106

97. For a reaction in which case the activation energies of forward and reverse reactions are equal :–
(1) $\Delta H = 0$ (2) $\Delta S = 0$
(3) The order is zero (4) There is no catalyst
CK0107

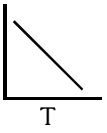
98. The energy of activation of a forward reaction is 50 kCal. The energy of activation of its backward reaction is:–
(1) Equal to 50 kCal.
(2) Greater than 50 kCal.
(3) Less than 50 kCal.
(4) Either greater or less than 50 kCal.

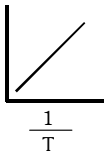
CK0108

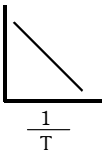
99. An exothermic reaction $X \rightarrow Y$ has an activation energy 30 kJ mol^{-1}. If energy change (ΔH) during the reaction is – 20 kJ, then the activation energy for the reverse reaction is :–
(1) 10 kJ (2) 20 kJ
(3) 50 kJ (4) – 30 kJ

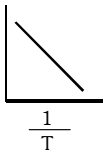
CK0109

100. Which of the following plot is in accordance with the arrhenius equation :–

(1) 

(2) 

(3) 

(4) 

CK0110

101. The rate of reaction increases by the increase of temperature because :–
(1) Collision is increased
(2) Energy of products decreases
(3) Fraction of molecules possessing energy $\geq E_T$ (Threshold energy) increases
(4) Mechanism of a reaction is changed

CK0111

102. For a certain gaseous reaction rise of temperature from 25° C to 35° C doubles the rate of reaction. What is the value of activation energy :–

(1) $\frac{10}{2.303R \times 298 \times 308}$ (2) $\frac{2.303 \times 10}{298 \times 308R}$

(3) $\frac{0.693R \times 10}{290 \times 308}$ (4) $\frac{0.693R \times 298 \times 308}{10}$

CK0112

103. For producing the effective collisions, the colliding molecules must posses:-
(1) A certain minimum amount of energy
(2) Energy equal to or greater than threshold energy
(3) Proper orientation
(4) Threshold energy as well as proper orientation of collision

CK0114

104. The half life for a reaction ..... of temperature:-
(1) Independent
(2) Increases with increase
(3) Decreases with increase
(4) Increases or decreases with increase

CK0115

105. The activation energy for a chemical reaction depends upon :-
(1) Temperature
(2) Nature of reacting species
(3) Concentration of the reacting species
(4) Collision frequency

CK0116

106. During decomposition of an activated complex
(1) Some energy is abosrbed
(2) Some energy is released
(3) No energy change
(4) Both (1) & (2)

CK0216

EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	3	3	1	4	1	1	2	4	2	4	3	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	2	1	3	2	2	4	3	3	2	2	3	4	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	2	3	2	1	2	1	2	1	3	4	3	1	3	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	4	1	2	2	4	4	4	3	2	3	4	3	2	3
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	1	3	2	3	3	2	4	3	1	2	2	4	2	1	2
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	4	4	3	3	2	3	3	3	2	1	3	3	4	4	2
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	3	3	3	4	4	4	1	4	3	3	3	4	4	3	2
Que.	106														
Ans.	2														

EXERCISE-III (Analytical Questions) Master Your Understanding

1. For the elementary reaction
$A + 2B \rightarrow$ Product,
the differential rate equation is :-

(1) $-\frac{1}{2}\frac{d[A]}{dt} = \frac{-d[B]}{dt} = k[A][B]^2$

(2) $-\frac{1}{2}\frac{d[A]}{dt} = \frac{d[B]}{dt} = k[A][B]^2$

(3) $\frac{-d[A]}{dt} = -\frac{1}{2}\frac{d[B]}{dt} = k[A][B]^2$

(4) $\frac{dA}{dt} = \frac{1}{2}\frac{d[B]}{dt} = k[A][B]^2$

CK0175

2. For the reaction $A \rightarrow B$ rate increases 5.19 times if the concentration of reactant A is tripled. The order of the reaction.

(1) –1 (2) 1.5 (3) 1 (4) 2

CK0176

3. For the non-stoichiometric reaction $2A + B \rightarrow C + D$, the following kinetic data were obtained in three different experiments, measured at 298 K.

Initial Conc. (A)	Initial Conc. (B)	Initial rate of formation of C ($mol\ L^{-1}s^{-1}$)
0.1M	0.1M	1.2×10^{-3}
0.1M	0.2M	1.2×10^{-3}
0.2M	0.1M	2.4×10^{-3}

(1) $rate = k[A][B]^2$ (2) $rate = k[A]$

(3) $rate = k[A][B]$ (4) $rate = k[A]^2[B]$

CK0177

4. The following mechanism has been proposed for the reaction of NO with Br_2 to form NOBr :

$NO(g) + Br_2(g) \rightleftharpoons NOBr_2(g)$

$NOBr_2(g) + NO(g) \rightarrow 2NOBr(g)$

If the second step is the rate determining step, the order of the reaction with respect to NO(g) is :

(1) 2 (2) 1

(3) 0 (4) 3

CK0178

5. Azo isopropane decomposes according to the equation :–

$(CH_3)_2CHN = NCH(CH_3)_2\ (g) \xrightarrow{250-290^0C} N_2(g) + C_6H_{14}(g)$

It is found to be a first order reaction. If initial pressure is P_o and pressure of the mixture at time t is P_t then rate constant k would be :–

(1) $k = \frac{2.303}{t}\log\frac{P_o}{2P_o - P_t}$

(2) $k = \frac{2.303}{t}\log\frac{P_o - P_t}{P_o}$

(3) $k = \frac{2.303}{t}\log\frac{P_o}{P_o - P_t}$

(4) $k = \frac{2.303}{t}\log\frac{2P_o}{2P_o - P_t}$

CK0181

6. Which of the following statement is false :-

(1) A fast reaction has a large rate constant and short half life

(2) For a first order reaction, successive half lives are equal

(3) For a first order reaction, the half life is independent of concentration

(4) The half life of a reaction is half the time required for the reaction to go for completion

CK0183

7. The reaction

$CH_3COOC_2H_5 + NaOH \rightarrow CH_3COONa + C_2H_5OH$ is :-

(1) Bimolecular reaction (2) II order reaction

(3) Both (1) & (2) (4) None

CK0184

8. Correct expression for the first order reaction is:-

(1) $C_t = C_0\ e^{k_1 t}$ (2) $C_t\ e^{k_1 t} = C_0$

(3) $\ell n\ \frac{C_0}{C_t} = -k_1 t$ (4) $\ell n\ \frac{C_t}{C_0} = k_1 t$

CK0185

9. The rate of first order reaction is 1.5×10^{-2} mol L^{-1} min^{-1} at 0.5 M concentration of the reactant. The half life of the reaction is

(1) 7.53 min (2) 0.383 min

(3) 23.1 min (4) 8.73 min

CK0186

10. In a first order reaction, the concentration of the reactant, decreases from 0.8 M to 0.4 M in 15 min. The time taken for the concentration to change from 0.1 M to 0.025 M is
(1) 7.5 min (2) 15 min
(3) 30 min (4) 60 min

CK0187

11. The rate equation for the reaction $2A + B \rightarrow C$ is found to be : rate= k[A][B]. The correct statement for this reaction is that the
(1) rate of formation of C is twice the rate of disappearance of A.
(2) Half life is a constant
(3) unit of k must be s^{-1}
(4) value of k is independent of the concentrations of A and B

CK0188

12. The half-life of a radio isotope is four hours. If the initial mass of the isotope was 200 g, the mass remaining after 24 h undecayed is :
(1) 3.125 g (2) 2.084 g
(3) 1.042 g (4) 4.167 g

CK0189

13. For a first order reaction $A \longrightarrow B$ the reaction rate at reactant concentration of 0.01 M is found to be 2.0×10^{-5} mol $L^{-1}s^{-1}$. The half life period of the reaction is :
(1) 220 s (2) 30 s (3) 300 s (4) 347 s

CK0190

14. For reaction $aA \rightarrow xP$. When [A] is 2.2 mM, the rate was found to be 2.4 mMs^{-1}. On reducing concentration of A to half, the rate changes to 0.6 mMs^{-1}. The order of reaction with respect to A is :
(1) 1.5 (2) 2.0
(3) 2.5 (4) 3.0

CK0191

15. The $t_{1/2}$ of a reaction is halved as the initial concentration of the reactant is doubled. What is the order of reaction ?
(1) 1 (2) 0
(3) 2 (4) 3

CK0192

16. The rate of a chemical reaction doubles for every 10°C rise of temperature. If the temperature is raised by 50°C, the rate of the reaction increases by about :-
(1) 32 times (2) 64 times
(3) 10 times (4) 24 times

CK0193

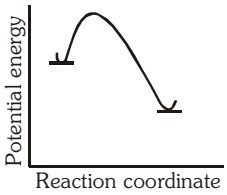
17. For a first order reaction, (A) $\rightarrow$ products, the concentration of A changes from 0.1 M to 0.025M in 40 min. The rate of reaction when the concentration of A is 0.01 M is :-
(1) 1.73×10^{-4} M/min (2) 1.73×10^{-5} M/min
(3) 3.47×10^{-4} M/min (4) 3.47×10^{-5} M/min

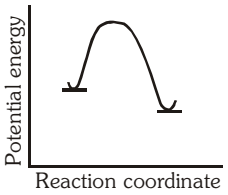
CK0194

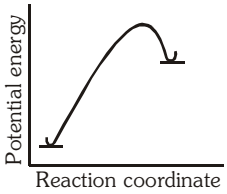
18. Arrhenius equation may be written as :–
(1) $\frac{d}{dT}(\ell n\ k) = -\frac{Ea}{RT}$ (2) $\frac{d}{dT}(\ell n\ k) = -\frac{Ea}{RT^2}$
(3) $\frac{d}{dT}(\ell n\ k) = +\frac{Ea}{RT^2}$ (4) $\frac{d}{dT}(\ell n\ k) = \frac{Ea}{RT}$

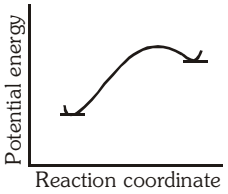
CK0195

19. An endothermic reaction with high activation energy for the forward reaction is given by the diagram :

(1) 

(2) 

(3) 

(4) 

CK0196

20. The rate of a reaction doubles when its temperature changes from 300 K to 310 K. Activation energy of such a reaction will be
(R = 8.314 JK^{-1} mol^{-1} and log 2 = 0.301)
(1) 53.6 kJ mol^{-1} (2) 48.6 kJ mol^{-1}
(3) 58.5 kJ mol^{-1} (4) 60.5 kJ mol^{-1}

CK0198

21. The rate constants k_1 and k_2 for two different reactions are $10^{16}\ e^{-2000/T}$ and $10^{15}\ e^{-1000/T}$, respectively.

The temperature at which $k_1 = k_2$ is

(1) $\frac{2000}{2.303}$K (2) 2000 K

(3) $\frac{1000}{2.303}$K (4) 1000 K

CK0119

22. The bromination of acetone that occurs in acid solution is represented by this equation

$CH_3COCH_3(aq)+Br_2(aq) \longrightarrow CH_3COCH_2Br(aq)+H^+(aq)+Br^-(aq)$

The kinetic data were obtained for given reaction at different concentrations.

$[CH_3COCH_3]$	$[Br_2]$	$[H^+]$	ROD of Br_2, Ms^{-1}
0.30	0.05	0.05	6×10^{-5}
0.30	0.10	0.05	6×10^{-5}
0.30	0.10	0.10	1.2×10^{-4}
0.40	0.05	0.20	3×10^{-4}

Based on these data, the rate equation is :

(1) Rate = $k[CH_3COCH_3]\ [Br_2]$

(2) Rate = $k[CH_3COCH_3]\ [Br_2][H^+]^2$

(3) Rate = $k[CH_3COCH_3]\ [Br_2][H^+]$

(4) Rate = $k[CH_3COCH_3]\ [H^+]$

CK0120

23. Rate of a reaction can be expressed as $r = PZ_{AB}e^{-Ea/RT}$. In the above expression which term is called steric factor and collision frequency of reaction A & B.

(1) Z_{AB} and $e^{-Ea/RT}$

(2) P and Z_{AB}

(3) Z_{AB} and $e^{-Ea/RT}$

(4) P and $e^{-Ea/RT}$

CK0221

24. For A(g) → B(g) + C(g) reaction the differential rate law expression is given as $r = \frac{-d[A]}{dt} = k[A]$.

Initially the pressure is 100 mm of Hg & after 10 minutes, total pressure is 120 mm of Hg. Hence rate constant (min^{-1}) is

(1) $k = \frac{2.303}{10}\log\frac{120}{100}$ (2) $k = \frac{2.303}{10}\log\frac{100}{20}$

(3) $k = \frac{2.303}{10}\log\frac{100}{80}$ (4) $k = \frac{2.303}{10}\log\frac{100}{120}$

CK0222

25. The rate constant for 1st order reaction is 69.3 s^{-1}. How much time will it take to reduce the initial concentration of the reactant to its $\frac{1}{16}$th value?

(1) 0.04 sec (2) 0.02 sec

(3) 0.06 sec (4) 0.05 sec

CK0223

26. For any chemical reaction, chemists try to find out

(1) the feasability of a chemical reaction which can be predicted by thermodynamics

(2) speed of a reaction

(3) extent to which a reaction will proceed can be determined

(4) All of the above

CK0224

27. $t_{1/2}$ for a zero order reaction is ….(I)…. proportional to the initial concentration of the reactants and ….(II)…. proportional to the rate constant. Choose the correct words to fill up the blanks (I) and (II) respectively from the options given below.

(1) (I) inversely, (II) directly

(2) (I) directly, (II) directly

(3) (I) inversely, (II) inversely

(4) (I) directly, (II) inversely

CK0225

28. What is true about pseudo first order reaction?

(1) One of the two reactants is in excess in the reaction

(2) There is only one reactant in the reaction

(3) It is actually a zero order reaction

(4) None of the above

CK0226

EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**2**	**2**	**1**	**1**	**4**	**3**	**2**	**3**	**3**	**4**	**1**	**4**	**2**	**3**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28		
Ans.	**1**	**3**	**3**	**3**	**1**	**3**	**4**	**2**	**3**	**1**	**4**	**4**	**1**		